Liam Shannon

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EDUCATION

Queen's University — **Kingston, Ontario**
BASc in Mechatronics and Robotics Engineering — *September 2024 – Present*

- **GPA:** 4.19/4.3
- **Related Coursework:** Data Structures & Algorithms, Robotics Design II, Computer Architecture, Signals & Systems

WORK EXPERIENCE

Cavtera — **Ottawa, Ontario / Remote**
Software Developer — *June 2026 – Present*

- Built React/OSDK apps for trucking operations, delivering custom software to clients across North America.
- Developed TypeScript functions within the Ontology SDK to power business logic across fleet workflows.
- Engineered Python data pipelines to cleanse and normalize thousands of rows of truck data, boosting productivity in trucking workflows by 50%.

Thomas Cavanagh Construction Limited — **Ottawa, Ontario / Remote**
Software Developer — *May 2025 – May 2026*

- Developed truck dispatch software in Palantir Foundry to manage the largest fleet in Eastern Ontario.
- Reduced compute costs by 200%+ on dispatch workflows through data pipeline optimization and querying logic.
- Worked hand-in-hand with employees across all divisions to create, iterate, and improve Foundry workflows.

Queen's Aerospace Design Team — **Kingston, Ontario**
Automation Manager — *July 2026 – Present*

- Managing the software stack, delegating tasks to team members, learning and teaching ROS 2 concepts.

System Integrations/ROS Simulation team member — *September 2024 – Present*

- Designed 3D CAD models in Onshape for integration on the team's test drone, improving structural fit.
- Developed and ran drone simulations in ROS to test flight, control and system integration in a virtual environment.

LEADERSHIP & VOLUNTEER EXPERIENCE

Math/Science Tutor — *September 2023 – Present*

- Tutored Calculus, Advanced Functions, Physics, and Chemistry at the University, and High School level.

Engineering Orientation Leader — *August 2025 – September 2025*

- Mentored incoming students, fostered an inclusive and engaging environment, and planned orientation events.

Ice Hockey Official — *September 2021 – February 2024*

- Officiated 100+ games, making quick, high-pressure decisions to uphold the integrity of the game.

PROJECTS

Autonomous Mars Rover Robot — *January 2026 – April 2026*

- Designed a fully autonomous four-wheeled Mars rover using ROS on a Raspberry Pi 4 for high-level decision making, sensor processing, and SLAM-based mapping and localization.
- Implemented path planning and autonomy in ROS, inducing velocity commands from localization and mapping data.
- Developed a low-level Arduino control system handling motor drivers, encoder feedback, and PID motor control for precise rover motion.

Line-Following Robot — *January 2025 – April 2025*

- Developed line following robot using an Arduino to program motor actions based on closed-loop sensor feedback.
- Tuned algorithms based on sensor data; optimized performance by integrating hardware and software subsystems.

SKILLS

Programming: TypeScript/JavaScript, React, Python, C/C++, NIOS II Assembly, VHDL

Design tools: SolidWorks, Onshape, Altium, LTspice